

Q1.

The four vertices of a parallelogram $ABCD$ have coordinates

$$A(1, 0, 2), B(3, -1, 5), C(7, 2, 4) \text{ and } D(5, 3, 1)$$

- (a) (i) Find $\overrightarrow{AB} \times \overrightarrow{AD}$. (3)
- (ii) Show that the area of the parallelogram is $p\sqrt{10}$, where p is an integer to be found. (2)
- (b) The diagonals AC and BD of the parallelogram meet at the point M . The line L passes through M and is perpendicular to the plane $ABCD$.
Find an equation for the line L , giving your answer in the form $(\mathbf{r} - \mathbf{u}) \times \mathbf{v} = \mathbf{0}$. (4)
- (c) The plane Π is parallel to the plane $ABCD$ and passes through the point $Q(6, 5, 17)$.
Find the coordinates of the point of intersection of the line L with the plane Π . (6)
- (Total 15 marks)

Q2.

A line and a plane have equations

$$\frac{x-3}{p} = \frac{y-q}{3} = \frac{z-1}{-1}$$

and

$$\mathbf{r} \cdot \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} = 10$$

respectively, where p and q are constants.

- (a) Show that the line is **not** perpendicular to the plane. (1)
- (b) In the case where the line lies in the plane, find the values of p and q . (4)
- (c) In the case where the angle, θ , between the line and the plane satisfies $\sin \theta = \frac{1}{\sqrt{6}}$, and the line intersects the plane at $z = 2$:

- (i) find the value of p ;

(5)

- (ii) find the value of q .

(2)

(Total 12 marks)

Q3.

The planes Π_1 and Π_2 have equations

$$\mathbf{r} \cdot \begin{bmatrix} 2 \\ 1 \\ 7 \end{bmatrix} = 10 \quad \text{and} \quad \mathbf{r} \cdot \begin{bmatrix} 3 \\ 1 \\ -4 \end{bmatrix} = 7$$

respectively.

- (a) Determine, to the nearest degree, the acute angle between Π_1 and Π_2 .

(4)

- (b) By setting $z = t$, find cartesian equations for the line of intersection of Π_1 and Π_2 in the form

$$\frac{x-a}{l} = \frac{y-b}{m} = z=t$$

(6)

- (c) The line L , with equation $\mathbf{r} = \begin{bmatrix} 20 \\ -1 \\ 7 \end{bmatrix} + \lambda \begin{bmatrix} 1 \\ 9 \\ 4 \end{bmatrix}$, intersects Π_1 at the point P and Π_2 at the point Q .

Show that $PQ = k\sqrt{2}$, where k is an integer.

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(6)

(Total 16 marks)

Q4.

The planes Π_1 and Π_2 have vector equations $\mathbf{r} \cdot \begin{bmatrix} 6 \\ 2 \\ 9 \end{bmatrix} = 5$ and $\mathbf{r} \cdot \begin{bmatrix} 10 \\ -1 \\ -11 \end{bmatrix} = 4$ respectively.

- (a) Write down cartesian equations for Π_1 and Π_2 .

(1)

- (b) Find a vector equation for the line of intersection of Π_1 and Π_2 .

(5)

- (c) The plane Π_3 has cartesian equation $5x + 3y + 11z = 28$.

Use your answer to part (b) to find the coordinates of the point of intersection of Π_1 ,

Π_2 and Π_3 .

(4)

- (d) Determine a vector equation for the plane which passes through the point (4, 1, 9) and which is perpendicular to both Π_1 and Π_2 .

(3)

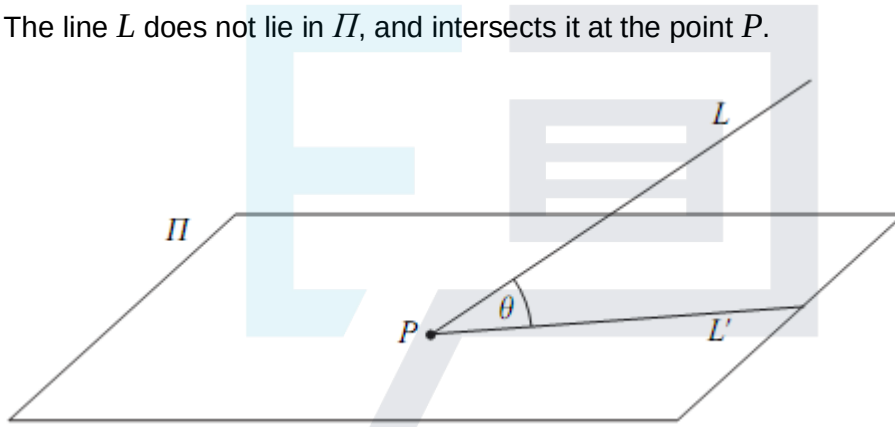
(Total 13 marks)

Q5.

The diagram shows the plane Π and the lines L and L' . The plane Π and the line L have equations

$$\mathbf{r} \cdot \begin{bmatrix} 3 \\ -2 \\ 6 \end{bmatrix} = 37 \quad \text{and} \quad \mathbf{r} = \begin{bmatrix} 1 \\ 2 \\ -7 \end{bmatrix} + \lambda \begin{bmatrix} 2 \\ 1 \\ 2 \end{bmatrix}$$

The line L does not lie in Π , and intersects it at the point P .



- (a) Determine the value of θ , the angle between L and Π , giving your answer to the nearest 0.1° .

(4)

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- (b) Find the coordinates of P .

(4)

- (c) The line L' lies in Π and is such that the angle between L and L' is θ , the angle between L and Π .

- (i) Find a vector which is parallel to Π and perpendicular to L .

(3)

- (ii) Hence, or otherwise, find a vector equation for L' in the form $\mathbf{r} = \mathbf{a} + \mu\mathbf{b}$.

(4)

(Total 16 marks)

Q6.

- (a) Find the value of p for which the planes with equations

$$\mathbf{r} \cdot \begin{bmatrix} 6 \\ -3 \\ 2 \end{bmatrix} = 42 \quad \text{and} \quad \mathbf{r} \cdot \begin{bmatrix} 4p+1 \\ p-2 \\ 1 \end{bmatrix} = -7$$

- (i) are perpendicular; (3)
- (ii) are parallel. (3)
- (b) In the case when $p = 4$:
- (i) write down a cartesian equation for each plane; (2)
- (ii) find, in the form $\mathbf{r} = \mathbf{a} + \lambda \mathbf{d}$, an equation for l , the line of intersection of the planes. (6)
- (c) Determine a vector equation, in the form $\mathbf{r} = \mathbf{u} + \beta \mathbf{v} + \gamma \mathbf{w}$, for the plane which contains l and which passes through the point $(30, 7, 30)$. (2)
- (Total 16 marks)

Q7.

The plane Π_1 is perpendicular to the vector $9\mathbf{i} - 8\mathbf{j} + 72\mathbf{k}$ and passes through the point $A(2, 10, 1)$.

- (a) Find, in the form $\mathbf{r} \cdot \mathbf{n} = d$, a vector equation for Π_1 . (3)
- (b) Determine the exact value of the cosine of the acute angle between Π_1 and the plane Π_2 with equation $\mathbf{r} \cdot (\mathbf{i} + \mathbf{j} + \mathbf{k}) = 11$. (4)

(Total 7 marks)

Q8.

The line L and the plane Π have vector equations

$$\mathbf{r} = \begin{bmatrix} 7 \\ 8 \\ 50 \end{bmatrix} + t \begin{bmatrix} 6 \\ 2 \\ -9 \end{bmatrix} \quad \text{and} \quad \mathbf{r} = \begin{bmatrix} -2 \\ 0 \\ -25 \end{bmatrix} + \lambda \begin{bmatrix} 5 \\ 3 \\ 4 \end{bmatrix} + \mu \begin{bmatrix} 1 \\ 6 \\ 2 \end{bmatrix}$$

respectively.

- (a) Show that L is perpendicular to Π . (3)
- (b) For the system of equations

$$\begin{aligned} 6p + 5q + r &= 9 \\ 2p + 3q + 6r &= 8 \\ -9p + 4q + 2r &= 75 \end{aligned}$$

form a pair of equations in p and q only, and hence find the unique solution of this system of equations.

(5)

(c) It is given that L meets Π at the point P .

(i) Demonstrate how the coordinates of P may be obtained from the system of equations in part (b).

(2)

(ii) Hence determine the coordinates of P .

(2)

(Total 12 marks)

Q9.

(a) The plane Π has equation $\mathbf{r} = \begin{bmatrix} 2 \\ 5 \\ 1 \end{bmatrix} + \lambda \begin{bmatrix} 0 \\ 3 \\ 1 \end{bmatrix} + \mu \begin{bmatrix} 4 \\ -1 \\ 0 \end{bmatrix}$.

(i) Find a vector which is perpendicular to both $\begin{bmatrix} 0 \\ 3 \\ 1 \end{bmatrix}$ and $\begin{bmatrix} 4 \\ -1 \\ 0 \end{bmatrix}$.

(2)

(ii) Hence find an equation for Π in the form $\mathbf{r} \cdot \mathbf{n} = d$.

(2)

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(b) The line L has equation $\left(\mathbf{r} - \begin{bmatrix} -1 \\ 2 \\ 6 \end{bmatrix} \right) \times \begin{bmatrix} 3 \\ 0 \\ -1 \end{bmatrix} = \mathbf{0}$.

Verify that $\mathbf{r} = \begin{bmatrix} 2 \\ 2 \\ 5 \end{bmatrix} + t \begin{bmatrix} 3 \\ 0 \\ -1 \end{bmatrix}$ is also an equation for L .

(2)

(c) Determine the position vector of the point of intersection of Π and L .

(4)

(Total 10 marks)